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EXPERIMENT – 3

Aim : To deploy multi-service applications using Docker Compose

LO Mapped : LO4

LO : Containerize, orchestrate and deploy applications using Docker, Docker Compose, Kubernetes and cloud infrastructure tools.

Theory :

Docker Compose

Docker Compose is a tool that helps in deploying and managing applications made up of multiple Docker containers. Rather than creating every container individually, all required services can be defined in a single configuration file called **docker-compose.yml**. This file contains the complete setup of the application, including the services, networks, ports, environment variables, and storage volumes.

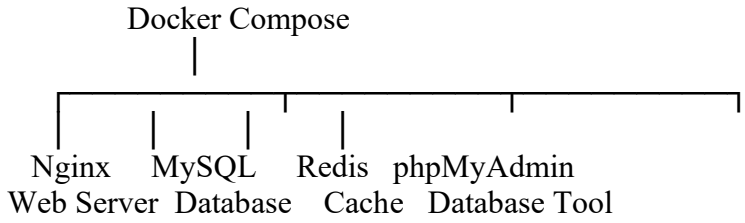
A single command is then sufficient to create and start the entire application:

docker compose up -d

The -d option starts the containers in detached mode so they continue running in the background while the terminal remains available for other commands.

Working Principle

Docker Compose reads the configuration from the **docker-compose.yml** file and automatically creates the required containers. It also establishes a private network through which all containers can communicate with each other using their service names instead of IP addresses.



Since every service is connected to the same Docker network, communication becomes simple. For instance, phpMyAdmin can connect to the MySQL service by using **database** as the hostname.

Implementation in the Experiment

This experiment demonstrated the deployment of a multi-container application using Docker Compose. Four different services were configured and executed together.

Nginx

Nginx served as the web server and was made accessible through:

http://localhost:8081

MySQL

MySQL acted as the database server. During deployment, a database named **studentdb** was automatically created for use by the application.

Redis

Redis was included as an in-memory data store. Its operation was verified using the Redis CLI by executing the **PING** command, which returned **PONG**, confirming that the service was functioning correctly.

phpMyAdmin

phpMyAdmin provided a browser-based interface for managing the MySQL database and was accessed through:

http://localhost:8082

docker-compose.yml :

services:

```
web:
  image: nginx
  container_name: webserver
  ports:
    - "8081:80"
database:
  image: mysql:8.0
  container_name: mysqlserver
  environment:
    MYSQL_ROOT_PASSWORD: root123
    MYSQL_DATABASE: studentdb
    MYSQL_USER: student
    MYSQL_PASSWORD: student123
  ports:
    - "3307:3306"
  volumes:
    - mysql-data:/var/lib/mysql
redis:
  image: redis:latest
  container_name: redisserver
  ports:
    - "6379:6379"
phpmyadmin:
```

```
    image: phpmyadmin:latest
    container_name: phpmyadmin
    environment:
      PMA_HOST: database
      PMA_PORT: 3306
      PMA_USER: student
      PMA_PASSWORD: student123
    ports:
      - "8082:80"
    depends_on:
      - database
volumes:
  mysql-data:
```

Common Docker Compose Commands:

Command	**Purpose**
mkdir compose-demo	Creates the project directory.
cd compose-demo	Opens the project folder.
notepad docker-compose.yml	Creates or edits the Compose configuration file.
dir	Displays the files present in the current directory.
docker compose config	Checks the configuration file for syntax errors.

Command	Purpose
docker compose up -d	Builds (if required) and starts all services in the background.
docker compose ps	Displays the status of all running Compose containers.
docker compose logs	Shows log messages generated by the services.
docker compose logs -f	Continuously streams container logs.
docker compose stop	Stops all running services without deleting them.
docker compose start	Starts the previously stopped containers.
docker compose restart	Restarts every service in the Compose application.
docker compose down	Stops and removes the containers along with the network.
docker compose down -v	Removes containers, networks, and associated volumes.
docker compose images	Lists the images used by the services.
docker exec -it redisserver redis-cli	Opens the Redis command-line interface.
docker ps	Displays all currently running Docker containers.

Execution :

```
C:\Users\Vineet Gajwani\compose-demo>docker compose config
no configuration file provided: not found
```

```
C:\Users\Vineet Gajwani>mkdir compose-demo

C:\Users\Vineet Gajwani>cd compose-demo

C:\Users\Vineet Gajwani\compose-demo>dir
 Volume in drive C is Windows-SSD
 Volume Serial Number is A01E-9EE5

 Directory of C:\Users\Vineet Gajwani\compose-demo

28-07-2026  19:19    <DIR>          .
28-07-2026  19:19    <DIR>          ..
               0 File(s)              0 bytes
               2 Dir(s)  789,670,744,064 bytes free
```

```
C:\Users\Vineet Gajwani\compose-demo>notepad docker-compose.yml

C:\Users\Vineet Gajwani\compose-demo>dir
 Volume in drive C is Windows-SSD
 Volume Serial Number is A01E-9EE5

 Directory of C:\Users\Vineet Gajwani\compose-demo

28-07-2026  19:24    <DIR>          .
28-07-2026  19:19    <DIR>          ..
28-07-2026  19:24               795 docker-compose.yml
               1 File(s)            795 bytes
               2 Dir(s)  789,670,215,680 bytes free

C:\Users\Vineet Gajwani\compose-demo>docker compose config
```

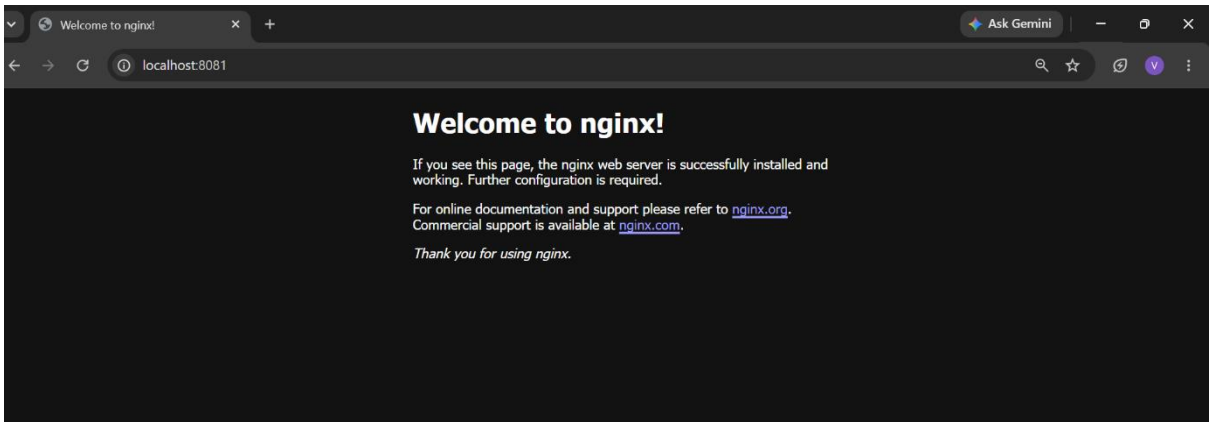
```
C:\Users\Vineet Gajwani\compose-demo>docker compose up -d
[+] Running 39/40
 ✔phpmyadmin Pulled                                                    51.3s
 ✔database Pulled                                                      63.1s
 ✔redis Pulled                                                         30.9s

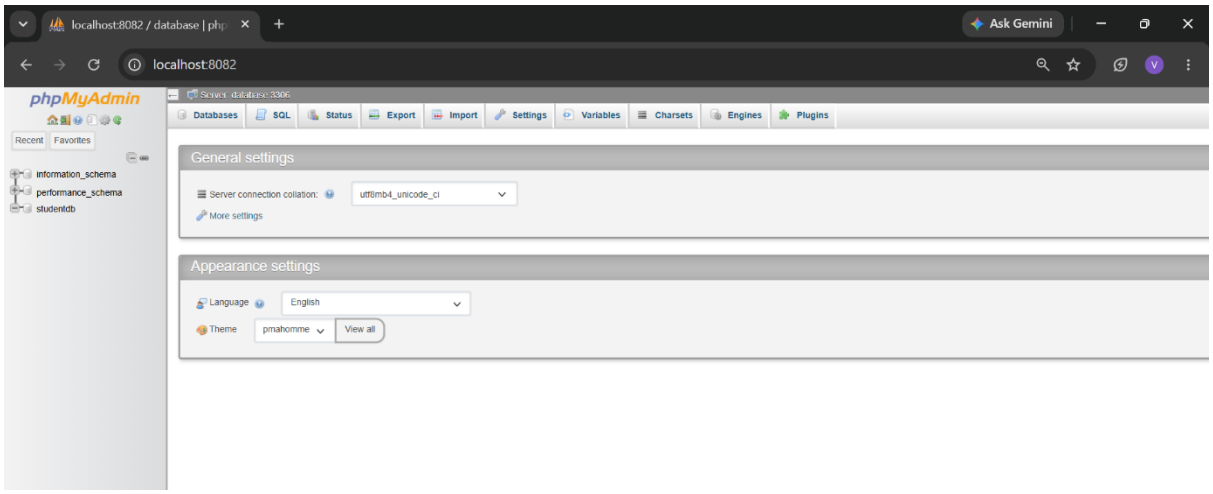



[+] Running 6/6
 ✔Network compose-demo_default     Created                              0.1s
 ✔Volume compose-demo_mysql-data   Created                              0.0s
 ✔Container redisserver            Started                              1.9s
 ✔Container mysqlserver            Started                              1.9s
 ✔Container webserver              Started                              1.9s
 ✔Container phpmyadmin             Started                              1.8s

C:\Users\Vineet Gajwani\compose-demo>
```

```
C:\Users\Vineet Gajwani\compose-demo>docker compose ps
NAME          IMAGE              COMMAND                  SERVICE      CREATED         STATUS         PORTS
mysqlserver   mysql:8.0          "docker-entrypoint.s…"   database     6 minutes ago   Up 6 minutes   0.0.0.0:3307->3306/tcp, [::]:3
307->3306/tcp
phpmyadmin    phpmyadmin:latest  "/docker-entrypoint.…"   phpmyadmin   6 minutes ago   Up 6 minutes   0.0.0.0:8082->80/tcp, [::]:808
2->80/tcp
redisserver   redis:latest       "docker-entrypoint.s…"   redis        6 minutes ago   Up 6 minutes   0.0.0.0:6379->6379/tcp, [::]:6
379->6379/tcp
webserver     nginx              "/docker-entrypoint.…"   web          6 minutes ago   Up 6 minutes   0.0.0.0:8081->80/tcp, [::]:808
1->80/tcp
```






```
C:\Users\Vineet Gajwani\compose-demo>docker exec -it mysqlserver mysql -u student -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 83
Server version: 8.0.46 MySQL Community Server - GPL

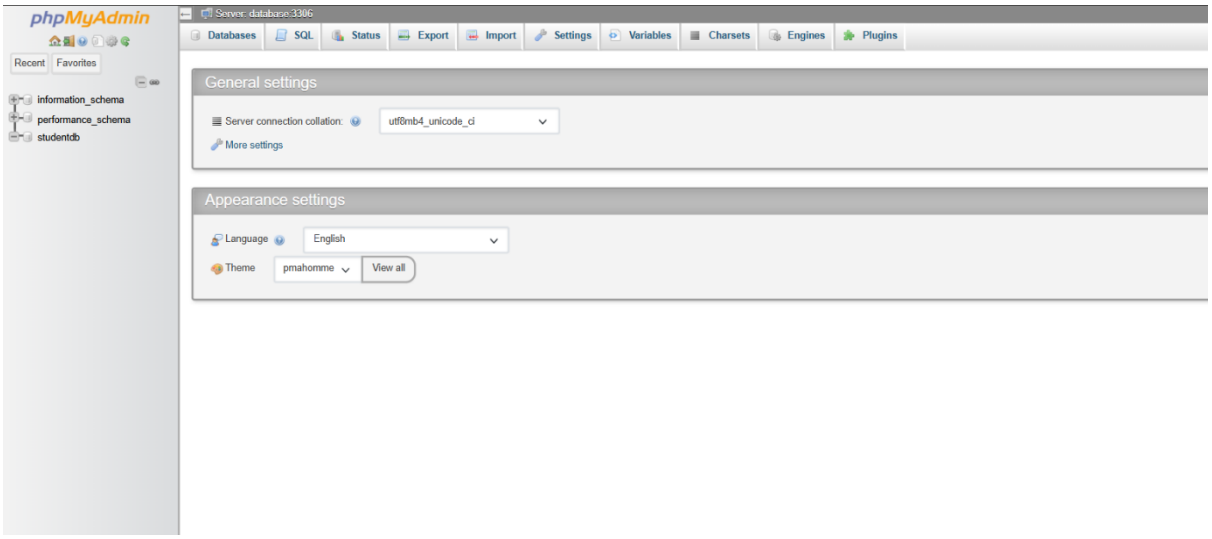
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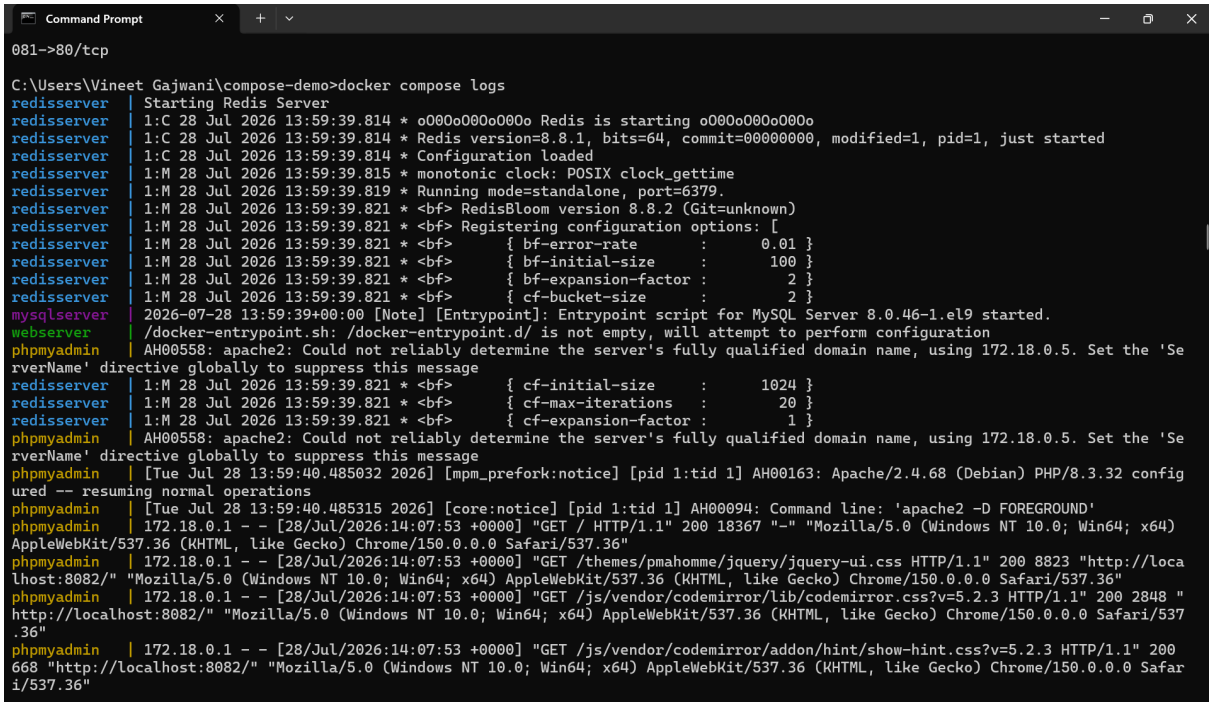
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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| performance_schema |
| studentdb          |
+--------------------+
3 rows in set (0.00 sec)
```

```
C:\Users\Vineet Gajwani\compose-demo>docker exec -it redisserver
 redis-cli
127.0.0.1:6379> PING
PONG
127.0.0.1:6379> exit
```






```
C:\Users\Vineet Gajwani\compose-demo>docker compose stop
[+] Stopping 4/4
 ✔Container redisserver  Stopped                0.7s
 ✔Container webserver    Stopped                0.6s
 ✔Container phpmyadmin   Stopped                1.5s
 ✔Container mysqlserver  Stopped                1.6s

C:\Users\Vineet Gajwani\compose-demo>docker compose ps
NAME      IMAGE     COMMAND   SERVICE   CREATED   STATUS    PORTS
```

```
C:\Users\Vineet Gajwani\compose-demo>docker compose start
[+] Running 4/4
 ✔Container redisserver  Started                                                0.6s
 ✔Container mysqlserver  Started                                                0.5s
 ✔Container webserver    Started                                                0.6s
 ✔Container phpmyadmin   Started                                                0.5s

C:\Users\Vineet Gajwani\compose-demo>docker compose ps
NAME          IMAGE              COMMAND                  SERVICE      CREATED          STATUS
 PORTS
mysqlserver   mysql:8.0          "docker-entrypoint.s…"   database     39 minutes ago   Up 12 seconds
 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp
phpmyadmin    phpmyadmin:latest  "/docker-entrypoint.…"   phpmyadmin   39 minutes ago   Up 12 seconds
 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
redisserver   redis:latest       "docker-entrypoint.s…"   redis        39 minutes ago   Up 12 seconds
 0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
webserver     nginx              "/docker-entrypoint.…"   web          39 minutes ago   Up 12 seconds
 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
```

```
C:\Users\Vineet Gajwani\compose-demo>docker compose restart
[+] Restarting 4/4
 ✔Container webserver    Started           0.9s
 ✔Container redisserver  Started           1.1s
 ✔Container mysqlserver  Started           2.0s
 ✔Container phpmyadmin   Started           2.0s

C:\Users\Vineet Gajwani\compose-demo>docker compose ps
NAME          IMAGE              COMMAND                  SERVICE      CREATED          STATUS
PORTS
mysqlserver   mysql:8.0          "docker-entrypoint.s…"   database     40 minutes ago   Up 5 seconds
0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp
phpmyadmin    phpmyadmin:latest  "/docker-entrypoint.…"   phpmyadmin   40 minutes ago   Up 5 seconds
0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
redisserver   redis:latest       "docker-entrypoint.s…"   redis        40 minutes ago   Up 6 seconds
0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
webserver     nginx              "/docker-entrypoint.…"   web          40 minutes ago   Up 6 seconds
0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
```

Conclusion :

Docker Compose simplified the process of deploying and managing multiple interconnected containers. All four services were configured within a single Compose file and their functionality was verified by performing operations such as starting, monitoring, restarting, and stopping the containers.